THE LOCUS WHERE A PERFECT COMPLEX IS A LOCALLY FREE MODULE OF RANK R IN DEGREE I IS OPEN AND CHARACTERIZED BY PULLBACK

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Allowed standard prerequisites (editorial).

- For a perfect complex the functions beta_j measuring the size of cohomology in each degree are upper semicontinuous, vanish outside a finite set locally, and their level sets interact well with pullback. invoked at proof lines 8029-8036. Provides the open locus W and its base change compatibility.
- A perfect complex whose cohomology is concentrated in a single degree is isomorphic to that cohomology sheaf shifted, and the sheaf is finite locally free when the complex is perfect. invoked at proof lines 8039-8044. Identifies E with H^i(E)[-i] on W, reducing to the rank stratification.

Source reference. Original lemma 0BDI: sources/stacks-perfect/pinned-original-source.tex, source lines 7907–7907. Cited at proof lines 8029-8036 to define the open W; the reduction to W and the rank stratification are inside the counted span.

Source reference. Original lemma 0BCD: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8039-8044 for E isomorphic to H^i(E)[-i]; the disjoint rank decomposition concluding the proof is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $E \in D(\mathcal{O}_X)$ be perfect. Given $i, r \in \mathbf{Z}$, there exists an open subscheme $U \subset X$ characterized by the following*

(1) *$E|_U \cong H^i(E|_U)[-i]$ and $H^i(E|_U)$ is a locally free $\mathcal{O}_U$-module of rank r,*
(2) *a morphism $f : Y \to X$ factors through U if and only if Lf^*E is isomorphic to a locally free module of rank r placed in degree i.*

Proof. Let $\beta_j : X \to \{0, 1, 2, \ldots\}$ for $j \in \mathbf{Z}$ be the functions of Lemma 0BDI. Then the set

$$W = \{x \in X \mid \beta_j(x) \leq 0 \text{ for all } j \neq i\}$$

is open in X and its formation commutes with pullback to any Y over X. This follows from the lemma using that apriori in a neighbourhood of any point only a

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finite number of the β_j are nonzero. Thus we may replace X by W and assume that $\beta_j(x) = 0$ for all $x \in X$ and all $j \neq i$. In this case $H^i(E)$ is a finite locally free module and $E \cong H^i(E)[-i]$, see for example More on Algebra, Lemma 0BCD. Thus X is the disjoint union of the open subschemes where the rank of $H^i(E)$ is fixed and we win. $\square$